waterschap amstel, gooi en vecht

By team Simon Simulations

AI FOR WATER HACKATHON · AGV CHALLENGE · PROTOTYPE, NOT AN OFFICIAL AGV PRODUCT

Better decisions on when to turn the pumps on and off.

Climate change makes the weather more extreme. AGV no longer only wants to get as much water away as possible: in warm, dry weather it also wants to **retain** water. We built two layers that work together, for all 43 polder pumping stations.

LAYER 1 · THE MODEL

Pumps the way AGV's operators do

A machine-learning model, trained on 26 years of AGV pump data and the weather, that does what the operators already do, by machine.

+

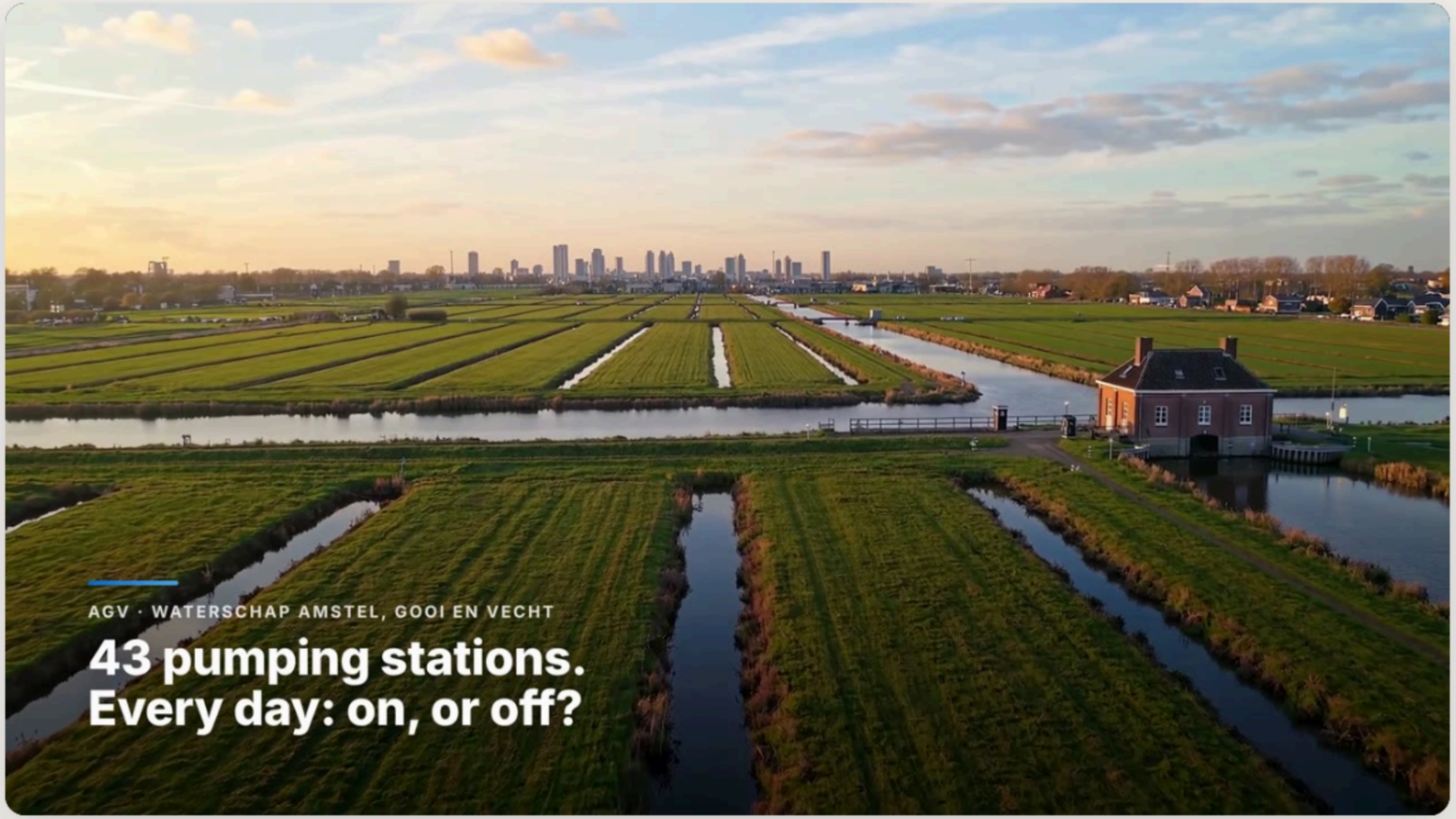
LAYER 2 · THE RULES

Decides when to deviate

Rules on top read the 14-day forecast: pump extra before heavy rain, pump less to retain water before a dry spell.

Disclaimer: this is a prototype built by a team of participants during the AI for Water Hackathon (Amsterdam, October 2026). It is not an official AGV product, and it is not made, endorsed or used by Waterschap Amstel, Gooi en Vecht. Results are estimates from a prototype.

THE IDEA IN ONE MINUTE



LAYER 1 · THE MODEL

A machine that pumps the way AGV's operators do.

1

Learned from AGV's own operation

Trained on 26 years (1996–2021) of hourly pump data from 43 stations, together with the real weather.

2

Tested on years it never saw

On 2022–2025 it matches the daily pumping of all stations together (R^2 **0.84**), and per station week by week (R^2 **0.93**).

3

One model, every station

It knows each polder's size, soil and land use, so it can say for any day: *this is what AGV would usually pump*.

That "usual" pumping is the baseline. The rules in layer 2 decide when, and how much, to deviate from it.

LAYER 2 · THE RULES ON TOP

Pump ahead of the rain, retain water ahead of the drought.

HEAVY RAIN COMING

Pre-pump

When heavy rain is forecast, we pump extra beforehand so the polder has room to take it.

DRY SPELL COMING

Retain

When a dry spell is forecast, we pump less and keep water in the polder for later.

AFTER A DRY MONTH

Keep the buffer

After a dry period, a bit of rain doesn't make us pump the saved water away.

ALWAYS

Stay within safe limits

Water levels and pumps always stay within the limits they're known to handle.

According to rough estimates, if AGV had used our system in these periods:

−22%

lower peak pumping in the September 2013 storm than AGV actually pumped

1.2 million m³

more water held in the polders at the driest point of spring 2023 than with usual pumping

0.9 million m³

more water held at the driest point of the 2018 drought than with usual pumping

Replays of the real weather with a noisy forecast. The data has no water levels, so levels are calculated: water not pumped stays in the polder's open water (ditches and lakes, from the TOP10NL map).

LIMITATIONS · NEXT STEPS

The rules are our assumptions. AGV's experience should set them.

LIMITATION

The rules are not validated yet

The thresholds on the previous slide (15 mm, −20 mm, 70% of the band) are our own choices. Water-board professionals need to check them, and they need testing with more data and real experiments.

LIMITATION

No water levels in the data

The data has pump volumes but no level sensors, so the polder levels are simulated from pumped volume and open-water area. Letting water in during droughts is not in the data either.

WHERE THE MODEL IS WEAKER

Single days at some stations

Per station, daily accuracy ranges from R^2 0.28 to 0.82; week by week it is much better (0.93). And a forecast beyond about a week is uncertain, so the rules act carefully on it.

NEXT STEP

Easy to improve with AGV's knowledge

The rules are a handful of readable numbers in one place. New data or experiments plug straight in: tune a rule with the operators, replay past years to test it, then trial it in manual mode at a few stations. The goal is smooth operation that anticipates the weather, without pumping water out only to let it back in. Finding the best rules takes experience, and that is not in this dataset or this hackathon.

HOW IT MEETS THE FOUR CRITERIA

Built for AGV's real problem, ready to use.

PROBLEM FIT

When to pump, and when to retain

AGV's question: optimise when the pumps go on and off, now that climate change means not only draining water, but sometimes keeping it in. The rules answer exactly that, for all 43 stations.

TECHNICAL QUALITY

Our own model, honestly tested

Trained on AGV's own data, checked on years it never saw. It matches how operators pump; the rules on top are simple and explainable.

IMPACT

Fewer floods, less drought

Pumping ahead of rain lowers the peak load on the shared main water; in dry spells the polders keep their water.

FEASIBILITY

Works with what AGV already has

Existing pump records and free weather forecasts, in a browser. Operators stay in charge: auto or manual mode.

Open the live demo →

A short guided tour, then the operator screen. No login needed.